that, we may compute the number of purchases where both appear, divided by the total number of purchases where at least Y appears. That would give us a better insight about what goes with what, we could define a threshold on the confidence for merging groups, and instead of picking a random item in the same group, showing the top 5 strongest associations.

Nevertheless, clustering items in groups will probably be a smart move, as it will help performance, allowing to run some algorithms on each group of items separately rather than on the whole catalog.

If you are interested in knowing more about non-personalized (and personalized) recommender systems, we suggest you to take a look at Practical Recommender Systems, a fine and thorough guide on the topic.

Back to our example, the gist is that we would need to start from this huge set of items, and partition them into disjoint groups. And of course, new items are added to the catalog all the time, and relations are dynamic, so we would need to be able to update both the list of items and the groups.

5.2 Reasoning on Solutions

In this and the following sections, we will mainly use the term "partition" to refer to the disjoint groups above. However, "group" and "set" can also appear as synonyms.

We will also restrict to the aggregative case, in other words two partitions can, at some point, be merged in a single,